TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM FOUR INTER ISLAMIC MOCK EXAMINATION

BIOLOGY 2A

(ACTUAL PRACTICAL)

33/2A

(For Both School and Private Candidates)

wazaelimu.com

TIME: 2:30 HOURS **Monday, 26th August 2024 a.m.**

Instructions

1. This paper consists of **two (2)** questions. Answer BOTH questions.
2. Each question carries **twenty-five (25)** marks.
3. All writing should be in blue or black ink except for diagrams that must be drawn in pencil.
4. All answers must be written in the booklet(s) provided.
5. All communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer booklet(s) provided.

1. You are provided with solid sample of potassium permanganate.
   (a) Put 200cm^3 of distilled water or tap water in the beaker and put crystal of potassium permanganate into the water.
   (b) Leave the mixture undisturbed for ten minutes.
   (c) Observe the changes after ten minutes.

   Questions:
   (a) Explain what you have observed
   (b) What conclusion can you draw from your observation?
   (c) Name the physiological process investigated in the experiment.
   (d) Outline three significances of the process named in (c) to plants and animals.
   (e) Write two organs in human being and plants that perform their work by the process named in (c) above.

2. You have been provided with specimens U, V, W and X. Study them carefully and then answer the questions that follow:
   (a) Identify the specimens u – x by giving their common names.
   (b) How would you differentiate specimen U from specimen V by using three observable features?
   (c) As a taxonomist classify the specimen W to class level.
   (d) Justify the statement that, "Specimen W should be preserved for the betterment of agricultural activities". Give three points.
   (e) As an agricultural officer, identify the type of germination that takes place in specimen X. Illustrate your answer.
   (f) Why specimens V are said to be the most successful species in the planet earth? Give four points.

Wabillah Tawfiq